



K. J. Somaiya College of Engineering, Mumbai-77
(Somaiya Vidyavihar University)

Batch: C3

Roll No.: 16010123325

Experiment No. 2

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

Title: Represent discrete time signals and perform different operations on them.

Objective: To familiarize the beginner to MATLAB by introducing the basic features and commands of the program.

Expected Outcome of Experiment:

CO	Outcome
CO1	Identify various discrete time signals and systems and perform signal manipulation

Books/ Journals/ Websites referred:

1. <http://www.mathworks.com/support/>
2. www.math.mtu.edu/~msgocken/intro/intro.html
3. www.mccormick.northwestern.edu/docs/efirst/matlab.pdf
4. A.Nagoor Kani "Digital Signal Processing", 2nd Edition, TMH Education.

Pre Lab/ Prior Concepts:

Using MATLAB we can easily generate all basic functions such as unit step, ramp, growing and decaying exponential, etc. The various signals plotted in this program are Step signal,



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Ramp signal, Exponential signal etc

1. Unit Step Signal

The step signal is defined as

$$U[n] = k ; \text{ if } n \geq 0 \\ = 0 ; \text{ otherwise}$$

When $k=1$ it is called as unit step signal.

2. Ramp Signal

The ramp signal is defined

$$r[n] = n ; \text{ if } n \geq 0 \\ = 0 ; \text{ otherwise}$$

3. Exponential Signal

The exponential signal is defined as

$$X[n] = a^n$$

When 'a' is greater than 1 it is **increasing** exponential

When 'a' is less than 1 it is **decaying** exponential.

4. Impulse Signal

The impulse signal is defined

$$d[n] = k ; \text{ if } n=0 \\ = 0 ; \text{ otherwise}$$

When $k=1$ it is called as unit impulse



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The functions used in this program are:

a. Ones

This function is used to create an array of all ones
Syntax: $Y = \text{ones}(m, n)$

Description:

$Y = \text{ones}(n)$ returns an n-by-n matrix of 1's.

An error message appears if n is not a scalar.

$Y = \text{ones}(m, n)$ or $Y = \text{ones}([m \ n])$ returns an m-by-n matrix of ones.

b. Zeros

This function is used to create an array of all zeros

Syntax: $Y = \text{zeros}(m, n)$

Description:

$Y = \text{zeros}(n)$ returns an n-by-n matrix of 0's.

An error message appears if n is not a scalar.

$Y = \text{zeros}(m, n)$ or $Y = \text{zeros}([m \ n])$ returns an m-by-n matrix of Zeros.

c. EXP

This function is used to plot exponential signals

Syntax: $Y = \exp(X)$

Description:

The exp function is an elementary function that operates element-wise on arrays. Its domain includes complex numbers.

$Y = \exp(X)$ returns the exponential for each element of X. For complex, it returns the complex exponential.



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Steps with Syntax for representation of above discrete time signals:

```
n = 0:10;
```

% 1. Impulse signal

```
subplot(3,3,1);  
x = 1.*(n == 0);  
stem(n, x, 'filled');  
title('Impulse Signal');  
xlabel('n');  
ylabel('x[n]');  
grid on;
```

% 2. Unit signal

```
subplot(3,3,2);  
x = 1.*(n >= 0);  
stem(n, x, 'filled');  
title('Unit Signal');  
xlabel('n');  
ylabel('x[n]');  
grid on;
```

% 3. Ramp signal

```
subplot(3,3,3);  
x = n;  
stem(n, x, 'filled');  
title('Ramp Signal');  
xlabel('n'); ylabel('x[n]');  
grid on;
```

```
a=0.9;
```

% 4. Exponential Signal

```
subplot(3,3,4);  
x = a.^n;  
stem(n, x, 'filled');  
title('Exponential a^n');  
xlabel('n');  
ylabel('x[n]');  
grid on;
```

```
subplot(3,3,5);  
x = a.^-n;  
stem(n, x, 'filled');  
title('Exponential a^{-n}');  
xlabel('n');  
ylabel('x[n]');  
grid on;
```



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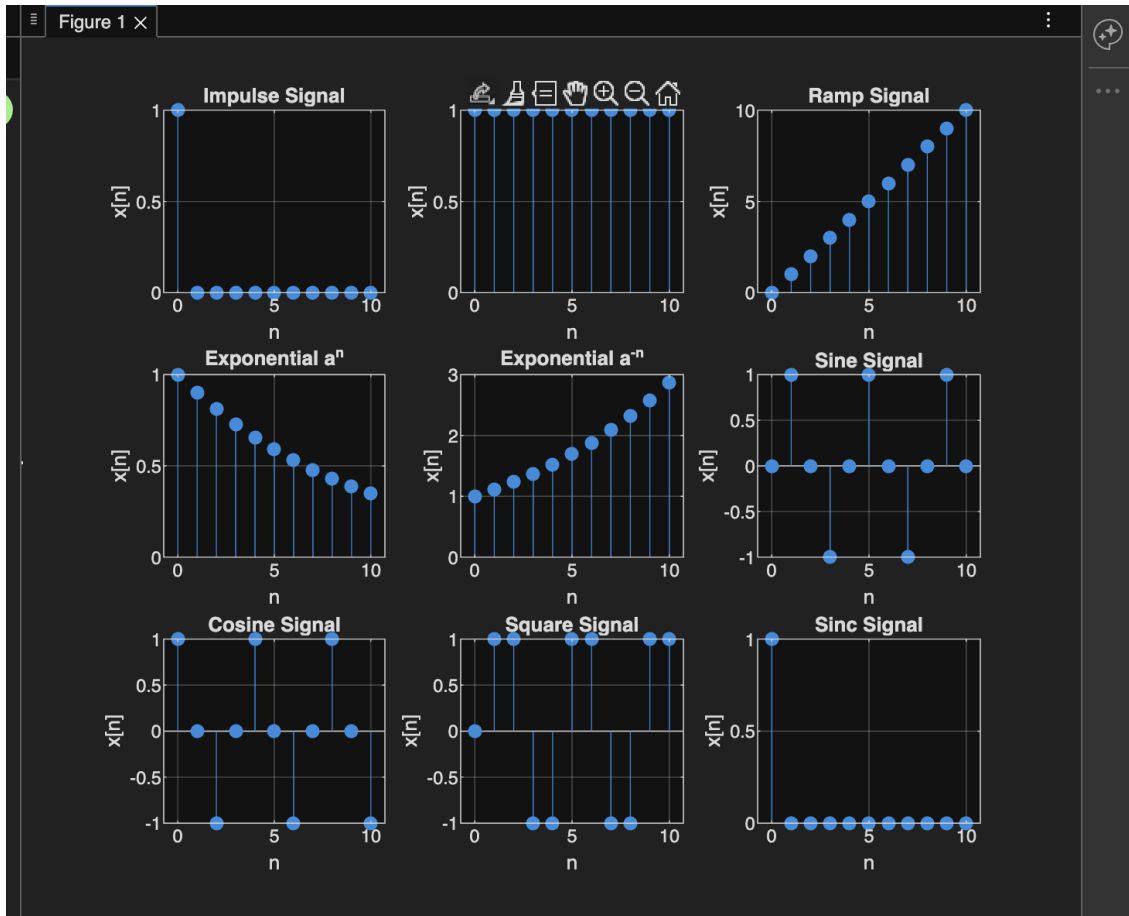
```
w=0.5*pi;  
% 6. Sine signal  
subplot(3,3,6);  
x = sin(w*n);  
stem(n, x, 'filled');  
title('Sine Signal');  
xlabel('n');  
ylabel('x[n]');  
grid on;
```

```
% 7. Cosine signal  
subplot(3,3,7);  
x = cos(w*n);  
stem(n, x, 'filled');  
title('Cosine Signal');  
xlabel('n'); ylabel('x[n]');  
grid on;
```

```
% 8. Square signal  
subplot(3,3,8);  
x = sign(sin(w*n));  
stem(n, x, 'filled');  
title('Square Signal');  
xlabel('n');  
ylabel('x[n]');  
grid on;
```

```
% 9. Sinc signal  
subplot(3,3,9);  
x = sinc(n);  
stem(n, x, 'filled');  
title('Sinc Signal');  
xlabel('n'); ylabel('x[n]');  
grid on;
```


Output:



Operations on Signals:

1. Addition of signals.
2. Subtraction of signals.
3. Multiplication of two signals.
4. Scaling – Upscaling & Downscaling.
5. Shift operation – Advance/Right shift & Delay/Left shift.



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Steps with Syntax for representation of above operations on discrete time signals:

main.m

```
n = -3:1:5;
```

```
y = @(n) n;
```

```
y0 = y(n);      % original
y1 = 1.5 * y(n); % amplified
y2 = 0.5 * y(n); % attenuated
y3 = y(n - 2);   % delayed
y4 = y(n + 2);   % advanced
y5 = y(-n-3);    % fold & advance
y6 = y(n.^2);    % squared input
y7 = y(-n + 3);  % fold & delay
y8 = y(-n);      % fold
```

plot.m

```
subplot(3,3,1);
```

```
stem(n, y0, 'filled');
```

```
title('Original Signal');
```

```
subplot(3,3,2);
```

```
stem(n, y1, 'filled');
```

```
title('Amplified Signal');
```

```
subplot(3,3,3);
```

```
stem(n, y2, 'filled');
```

```
title('Attenuated Signal');
```

```
subplot(3,3,4);
```

```
stem(n, y3, 'filled');
```

```
title('Delayed Signal');
```

```
subplot(3,3,5);
```

```
stem(n, y4, 'filled');
```

```
title('Advanced Signal');
```

```
subplot(3,3,6);
```

```
stem(n, y5, 'filled');
```

```
title('Fold & Advance');
```

```
subplot(3,3,7);
```

```
stem(n, y6, 'filled');
```

```
title('Squared');
```

```
subplot(3,3,8);
```

```
stem(n, y7, 'filled');
```

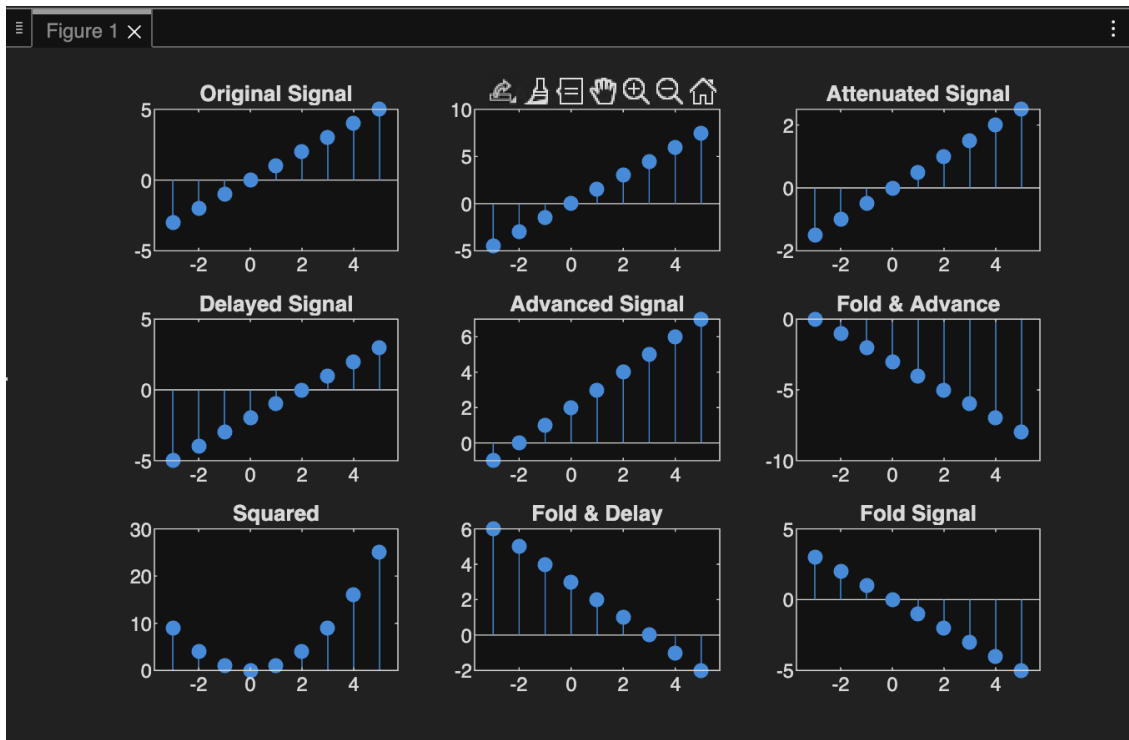
```
title('Fold & Delay');
```

```
subplot(3,3,9);  
stem(n, y8, 'filled');  
title('Fold Signal');
```



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Output:



Conclusion:

The above experiment deals with plotting of standard discrete signals in a 3x3 subplot and then performing standard operations on signals using predefined functions in MATLAB.



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Post Lab Questions

1. Let $x(n) = 8(0.5)^n (u[n+1] - u[n-3])$. Sketch the following signals

I. $Y(n) = [x-3]$

II. $F(n) = x[n+1]$

III. $G(n) = x[-n+4]$

$n = -5:5;$

$x1 = 8*(0.5).^n;$ % exponential part
 $x2 = 1.*((n >= -1) \& (n <= 2));$ % window using unit steps

$x = x1 .* x2;$

$y = 8*(0.5).^{(n-3)} .* (((n-3) >= -1) \& ((n-3) <= 2));$
 $f = 8*(0.5).^{(n+1)} .* (((n+1) >= -1) \& ((n+1) <= 2));$
 $g = 8*(0.5).^{(-n+4)} .* (((-n+4) >= -1) \& ((-n+4) <= 2));$

$\text{subplot}(2,2,1);$
 $\text{stem}(n,x);$
 $\text{title}('x(n)');$

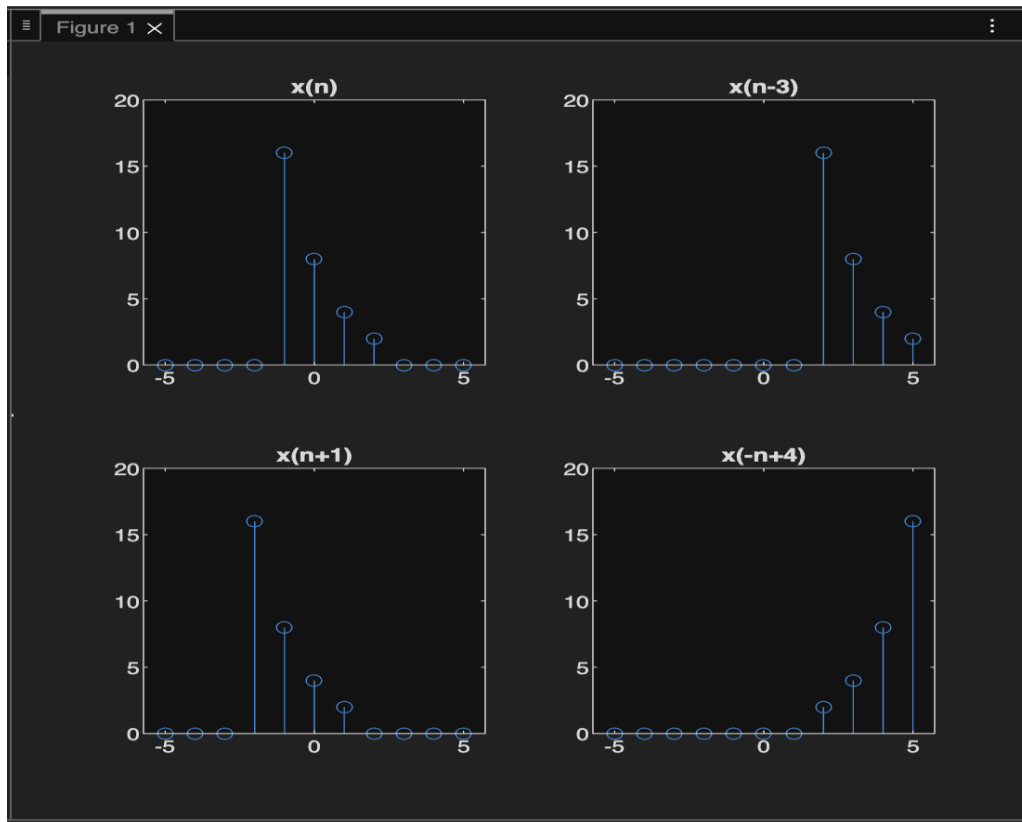
$\text{subplot}(2,2,2);$
 $\text{stem}(n,y);$
 $\text{title}('x(n-3)');$

$\text{subplot}(2,2,3);$
 $\text{stem}(n,f);$
 $\text{title}('x(n+1)');$

$\text{subplot}(2,2,4);$
 $\text{stem}(n,g);$
 $\text{title}('x(-n+4)');$



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2. The process of conversion of continuous time signal into discrete time signal is known as ._____ Sampling_____

3. Which of the following is example of deterministic signal?

- a. Step
- b. Ramp
- c. Exponential
- d. All of the above



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4. For energy signals the energy will be finite and the average power will be ___zero.

5. In a signal $x(n)$, if 'n' is replaced by 'n/3' the it is called. ___Expansion___.

6. The system $y(n)=\sin[x(n)]$ is

- a. Stable
- b. BIBO stable**
- c. Unstable
- d. None of the above